CS 6515-Exam 3
Spring 2026

Name: _____
GTID: _____

No notes, calculators, or other aids are allowed. Read all directions carefully and write your answers in the space provided.
Taking this exam signifies you are aware of and in accordance with the Academic Honor Code of Georgia.
Do not separate any pages from the rest of your exam.

PRACTICE EXAM

1. **You do not need to show work for any of these questions.**

[1] (i) Consider the Primal LP

$$\begin{aligned} \max\ & c^T x \\ s.t.:\ & Ax \leq b \\ & x \geq 0 \end{aligned}$$

You are given two vectors x, in the feasible region of the Primal, and y in the feasible region of the Dual LP. Which of the following statements are true? Check ALL that apply.

- ◯ $c^T x \leq b^T y$.
- ◯ $Ay \geq c$.
- ◯ The Primal LP is bounded.
- ◯ The Dual LP maximizes the linear function $b^T y$.

[1] (ii) If the Dual LP is infeasible. Which of the following can hold for the Primal LP? Check ALL that apply.

- ◯ Primal is feasible and has a bounded optimum.
- ◯ Primal LP is feasible and has unbounded optimum.
- ◯ Primal LP is infeasible.

[1] (iii) Consider the Primal LP

$$\begin{aligned} \max\ & c^T x \\ s.t.:\ & Ax \leq b \\ & x \geq 0 \\ & x \in \mathbb{R}^n \end{aligned}$$

You are told this input is feasible and bounded.

True or False: if the Primal LP has more than one optimal solution, then it has infinitely many.

- ◯ True.
- ◯ False.

[1] (iv) A 5-approximation algorithm is applied to a maximization problem, and the optimal solution is known to be 20. The approximation algorithm could return a value of 3:

- ◯ True.
- ◯ False.

[5] 2. (i) Write the Dual LP of the following

$$\begin{aligned} \min \ & 2x_2 - 5x_3 \\ s.t.: \ & 2x_1 + x_2 \geq 4 \\ & x_1 - 2x_2 - 3x_3 \leq 4 \\ & x_1 + x_2 = 10 \\ & x_i \geq 0 \end{aligned}$$

[5] (ii) Consider the following Greedy Approximation Algorithm for `Knapsack`.

Algorithm 1 (A greedy 2-Approx for 0/1 Knapsack)

Sort items such that $\frac{v_1}{w_1} \geq \frac{v_2}{w_2} \geq \cdots \geq \frac{v_n}{w_n}$
$S \leftarrow \emptyset$, $W_{rem} \leftarrow W$
for $i = 1$ **to** n **do**
  if $w_i \leq W_{rem}$ **then**
    $S \leftarrow S \cup \{i\}$, $W_{rem} \leftarrow W_{rem} - w_i$
  end if
end for
$i^* \leftarrow \text{argmax}\{v_i \mid w_i \leq W\}$
return $\text{argmax}_{S,\{i^*\}}\{\text{value}(S), v_{i^*}\}$

Informally, we sort the items by their *density* and fill our knapsack according to this sorting. Lastly, we select the largest between our value above and the most valuable item that fits in the knapsack.

Show that this is a 2-approximation of `Knapsack`.

[10] 3. Consider the `Maximum Independent Set` problem: given a graph $G = (V, E)$, we wish to find the largest subset $S \subseteq V$ such that

$$\forall u, v \in S \mid (uv) \notin E$$

(a) Design a Greedy Algorithm to approximate `MIS`.

(b) Assume your input graph has max degree Δ. Show that your design achieves a $(\Delta + 1)-$approximation. (*Hint: how many vertices you remove in every step?*)